🛠️ Network Troubleshooting in AWS

Tools like `curl`, `netstat`, `ss`, `telnet`, `ping`, and `traceroute` are essential for identifying network problems within your AWS cloud infrastructure. They help you diagnose issues related to:

- ☑ Connectivity
- ☑ Port availability
- ☑ DNS resolution
- ☑ Routing and latency

🔄 1. curl

- **Purpose**: Tests HTTP/HTTPS connectivity.
- **Use case**: Check if a web server or API endpoint is reachable.
- **Useful for**:
  - Verifying if a service is running.
  - Confirming DNS resolution works.
  - Detecting security group/firewall blocking HTTP traffic.

Example:
'curl http://example.com'

🌐 2. netstat and ss

- **Purpose**: Show network connections, open/listening ports, and routing tables.
- **Use case**: See if a service is listening or if connections are being made.
- **Useful for**:
  - Checking if applications are listening on correct ports.
  - Identifying failed, established, or closed connections.

Examples:
'ss -tuln'
'netstat -an'

📞 3. telnet

- **Purpose**: Tests if a specific port on a remote host is reachable.
- **Use case**: Check if your EC2 can reach another service, like a database.
- **Useful for**:
  - Verifying port access through security groups or NACLs.

Example:
'telnet 10.0.0.12 80'

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4. ping

- **Purpose**: Tests basic IP connectivity using ICMP.
- **Use case**: Confirm if another server or instance is reachable.
- **Useful for**:
    - Detecting basic network reachability issues.

Note: Some AWS services (like RDS) block ICMP by default.

Example:
'ping 10.0.0.12'

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5. traceroute

- **Purpose**: Shows the path packets take to reach a destination.
- **Use case**: Identify where latency or network failure occurs.
- **Useful for**:
    - Diagnosing routing issues inside and outside your VPC.

Example:
'traceroute google.com'

AWS-Specific Troubleshooting Use Cases

These tools help detect:

Problem	Detected By
Blocked ports or wrong SG/NACL configs	telnet, ss, netstat
Service not running	ss, netstat, curl
DNS resolution issues	curl, ping
Packet loss or latency	ping, traceroute
Misconfigured routing/IGW	traceroute, ping
